

PF Lab 12 Tasks (5)

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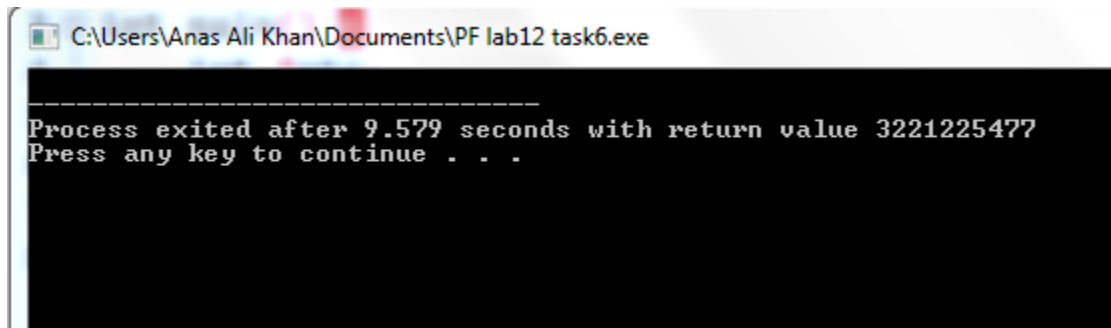
Section: BCS-1K

Roll No: 25K-0516

Task 6

```
#include <stdio.h>
```

```
int main() {  
    int *ptr;  
    *ptr = 10;  
    printf("Value: %d\n", *ptr);  
    return 0;  
}
```



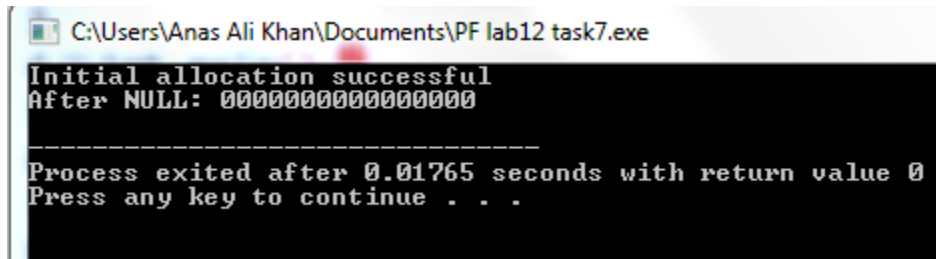
Task 7

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {  
    int *ptr=(int*)malloc(5 * sizeof(int));  
    printf("Initial allocation successful\n");  
  
    ptr = NULL;  
    printf("After NULL: %p\n", (void*)ptr);  
}
```

```
free(ptr);  
  
return 0;  
  
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab12 task7.exe  
Initial allocation successful  
After NULL: 0000000000000000  
-----  
Process exited after 0.01765 seconds with return value 0  
Press any key to continue . . .
```

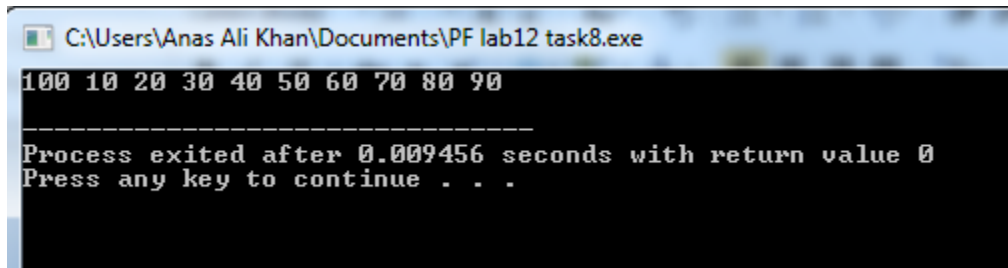
Task 8

```
#include <stdio.h>  
  
#include <stdlib.h>  
  
int main() {  
  
    int *arr=(int*)malloc(10 * sizeof(int));  
  
  
    for(int i = 0; i < 10; i++) {  
  
        arr[i] = i * 10;  
  
    }  
  
  
    arr[0] = 100;  
  
  
    for(int i = 0; i < 10; i++) {  
  
        printf("%d ", arr[i]);  
  
    }  
  
}
```

```
printf("\n");

return 0;

}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab12 task8.exe
100 10 20 30 40 50 60 70 80 90
-----
Process exited after 0.009456 seconds with return value 0
Press any key to continue . . .
```

Task 9

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int n;
```

```
    printf("Enter number of integers: ");
```

```
    scanf("%d", &n);
```

```
    int *arr = (int *)malloc(n);
```

```
    for(int i = 0; i < n; i++) {
```

```
        arr[i] = i * 5;
```

```
    }
```

```
    for(int i = 0; i < n; i++) {
```

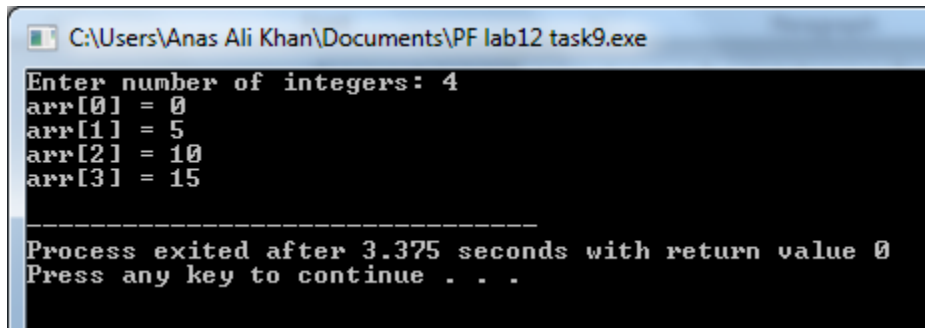
```

        printf("arr[%d] = %d\n", i, arr[i]);
    }

    free(arr);

    return 0;
}

```



```

C:\Users\Anas Ali Khan\Documents\PF lab12 task9.exe
Enter number of integers: 4
arr[0] = 0
arr[1] = 5
arr[2] = 10
arr[3] = 15
-----
Process exited after 3.375 seconds with return value 0
Press any key to continue . . .

```

Task 10

```

#include <stdio.h>

#include <stdlib.h>

int main() {

    int *ptr = (int *)malloc(5 * sizeof(int));

    int *new_ptr = (int *)realloc(ptr, 10 * sizeof(int));

    free(ptr);

    ptr[0] = 100;

    printf("Value at old pointer: %d\n", ptr[0]);
}

```

```
free(new_ptr);  
  
return 0;  
  
}
```

